SafeNet AI

Problem Statement — SafeNet AI

In today's world, public places such as railway stations, college campuses, shopping malls, hospitals, parks, and city roads face increasing safety and security threats including violence, theft, suspicious activities, crowd panic, and emergencies. Although large amounts of data are continuously generated through CCTV cameras, microphones, social media platforms, and crime databases, existing public safety systems analyze these data sources separately and independently.

Traditional surveillance systems mainly depend on single-source monitoring such as CCTV-only observation or simple alarm-based detection. These systems are limited because they cannot understand the complete situation happening in real time. For example, a CCTV camera may detect a crowd, but it cannot hear panic sounds or understand threatening social media posts related to the same location. Similarly, audio systems cannot visually identify suspicious movement, and crime databases cannot analyze live environmental changes.

Due to this disconnected analysis, many warning signs are missed before dangerous incidents occur. Existing systems are also reactive rather than predictive, meaning they generate alerts only after an incident has already happened. In addition, current systems often produce a large number of false alarms and do not provide proper explanations for why an alert was triggered, reducing operator trust and response efficiency.

To address these limitations, this project proposes **SafeNet AI**, an Explainable Multimodal Public Safety Intelligence System using Machine Learning. The system combines multiple data sources such as CCTV video feeds, audio signals, social media text, and historical crime records to detect and predict public safety threats in real time.

By using Computer Vision, Audio Classification, NLP, Machine Learning, and Explainable AI techniques, the proposed system generates a unified risk score for each monitored location and explains the reason behind every alert. This helps security personnel identify high-risk situations earlier, reduce false alarms, improve response time, and make smarter public safety decisions.